

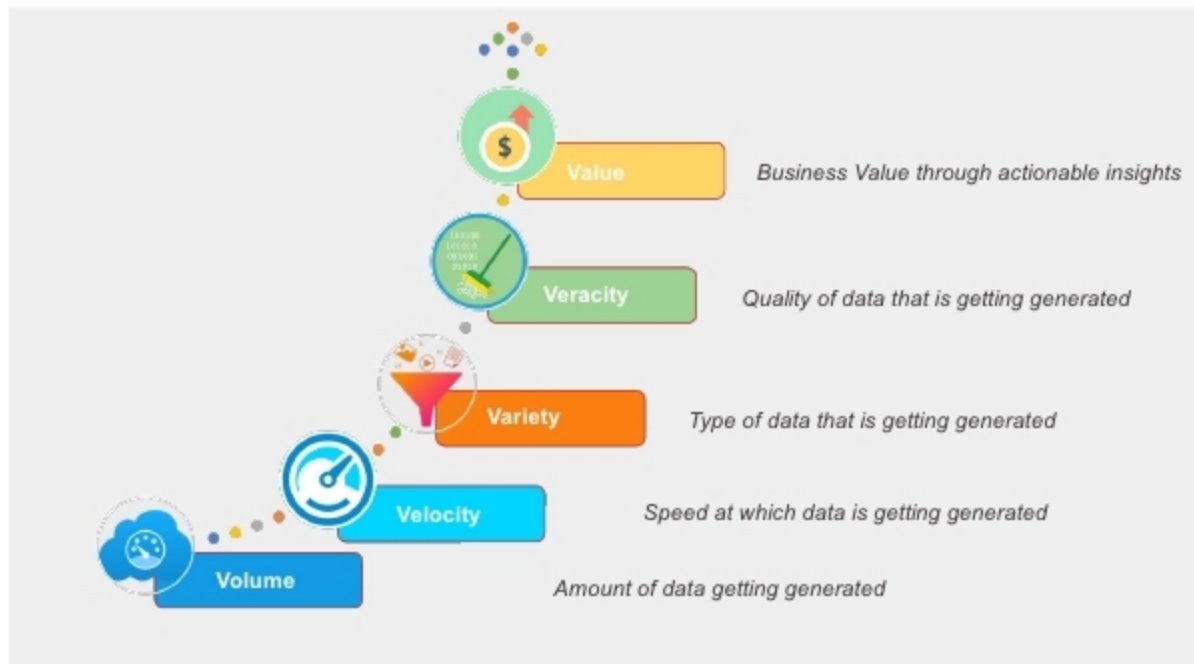


Figure 1: The Five Vs

Velocity: This refers to the speed at which data is getting generated. The need to capture and act on the data generated by millions of sensors, mobile devices, social media or financial transactions in real-time or near real-time has become unavoidable. Big Data technology can be leveraged to extract insights from the data streams itself, instead of having to save it in some database.

Variety: This refers to the different types of data that we can leverage to extract insights. Most of the data analysis in the pre-Big Data era was confined to structured data based on relational database tables. With Big Data technology, we can perform analysis on unstructured data, semi-structured data, quasi-structured data as well as structured data. Around 90 per cent of the current world data is unstructured and comes in formats as diverse as images, audio clips and videos, sensor data, social media data, logs, clickstream data and so on.

Veracity: This refers to the quality of data. To what extent is the data reliable and dependable? With different varieties of Big Data, it is impractical to control the data quality or expect that it is always reliable and trustworthy. While verifying the veracity of the data, we begin to understand the perils related with using the data for analysis and making business decisions based on it. The veracity of data also refers to how

meaningful it would be to leverage the context in which the data was generated, for analysis.

Value: This refers to the ability to extract business value out of the tsunami of data. It is imperative for enterprises to clearly understand the value generated by Big Data and the measurable outcomes that it would result in. Significant value can be found in Big Data including a 360° view of the customer, personalised targeting, operational efficiency, accelerated innovation and so on.

These characteristics of Big Data and the challenges associated with managing it warrant the use of new technical architectures and analytics to extract insights and enable new sources of revenue. This new technical architecture is named Hadoop and it can be considered a life saver.

Hadoop: A panacea for Big Data challenges

For most enterprises, data related initiatives are ranked at the top of the list when it comes to investments. One of the main reasons for this is that enterprises seek to unlock value from the huge amounts of structured, quasi-structured, semi-structured and unstructured data they have at their disposal to dramatically improve business outcomes. Hadoop is an innovative, interesting and powerful framework that can even dwarf Big

Data. Hadoop can be of immense help in many ways.

- **Capturing and storing all enterprise data:** A majority of enterprises estimate that they leverage less than one-fifth of the data they have for analysis, leaving most of their data unused. This is primarily because of the lack of analytics capabilities and ever-increasing data silos. Without analysis capabilities, it is very difficult to judge which data is going to be valuable and which is not. Data that may seem immaterial now might be a future gold mine. The cost and effort involved in managing data often results in enterprises having to make choices on what data to keep for analysis and what gets discarded. Thanks to Hadoop, enterprises now have an option to manage and analyse as much data as they want in a much more cost-effective way.
- **Rapid processing:** In traditional systems, data is shipped to the processing unit. This process is IO intensive and involves network hops as well. Hadoop uses a drastically different paradigm by leveraging data locality and distributed computing. In Hadoop, the processing is shipped to the data, thereby reducing the IO and network latency. In addition, processing is split into smaller tasks and multiple nodes work in parallel to complete these tasks very fast.
- **Democratising analytics:** Traditionally, analytics has been considered niche and has been restricted to a few strategic power users while tactical and operational users were deprived of it. The Hadoop ecosystem helps to democratise analytics by making it available to all users who make critical decisions at any level within an enterprise. In addition to distributed storage, Hadoop also provides distributed computing, which enables rapid crunching of large data volumes in parallel and at a swift pace.